QUASI-COMPACTNESS OF A SPACE IS TESTED ON ITS CLOSED POINTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source line 2024). Makes the complement of the lifted covering quasi-compact.
- The statement that a nonempty quasi-compact space with the standing hypotheses has a closed point. Invoked inside the counted proof (source lines 2025-2028). Produces the closed point whose existence contradicts the covering.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-closed-point: sources/stacks-topology/pinned-original-source.tex, source lines 1985–1985. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-closed-point, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a quasi-compact Kolmogorov space. Then the set X_0 of closed points of X is quasi-compact.*

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $X_0 = \bigcup U_{i,0}$ be an open covering. Write $U_{i,0} = X_0 \cap U_i$ for some open $U_i \subset X$. Consider the complement Z of $\bigcup U_i$. This is a closed subset of X, hence quasi-compact (Lemma lemma-closed-in-quasi-compact) and Kolmogorov. By Lemma lemma-quasi-compact-closed-point if Z is nonempty it would have a closed point which contradicts the fact that $X_0 \subset \bigcup U_i$. Hence $Z = \emptyset$ and $X = \bigcup U_i$. Since X is quasi-compact this covering has a finite subcover and we conclude. $\square$